Exp.No 2: Implementing TDMA Slot Allocation in GSM.

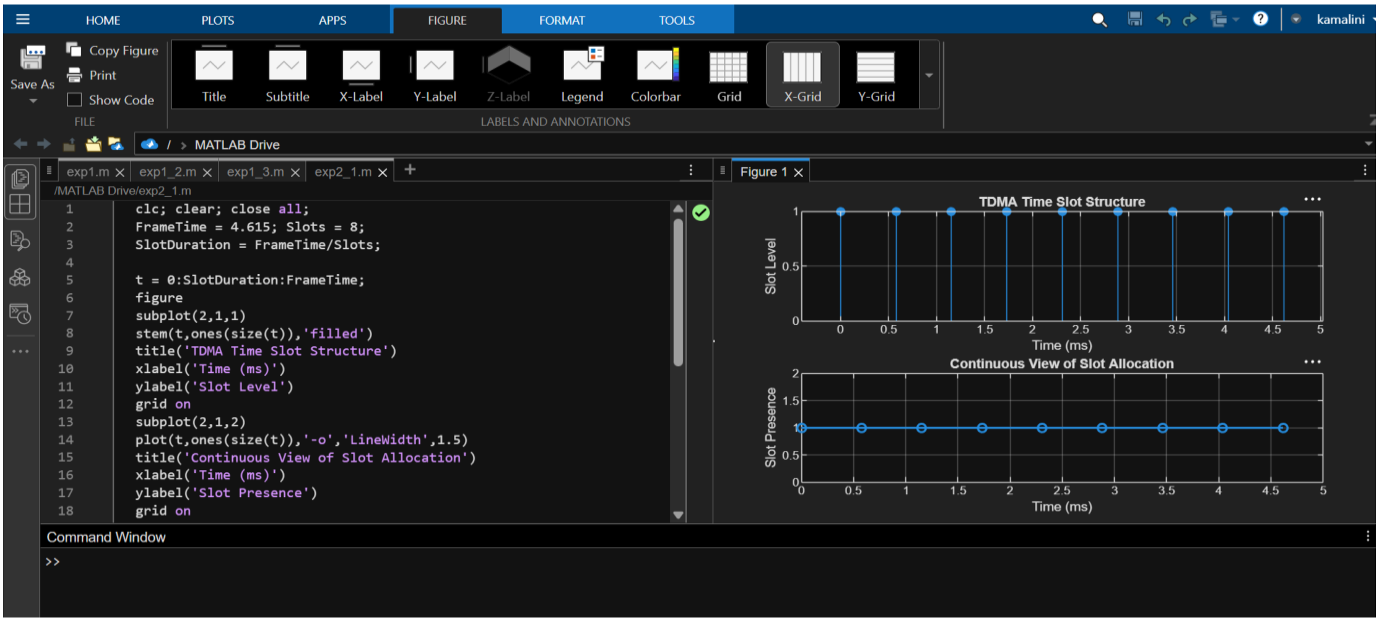
OBJECTIVE 1 :

1. To calculate the Time Slot Duration in a GSM TDMA frame by dividing the total frame time into equal time slots. This helps in understanding the time structure of GSM

CODE:

```
clc; clear; close all;
FrameTime = 4.615; Slots = 8;
SlotDuration = FrameTime/Slots;
t = 0:SlotDuration:FrameTime;
figure
subplot(2,1,1)
stem(t,ones(size(t)),'filled')
title('TDMA Time Slot Structure')
xlabel('Time (ms)')
ylabel('Slot Level'), subplot(2,1,2)
plot(t,ones(size(t)),'-o','LineWidth',1.5)
title('Continuous View of Slot Allocation')
xlabel('Time (ms)')
ylabel('Slot Presence')
grid on
```

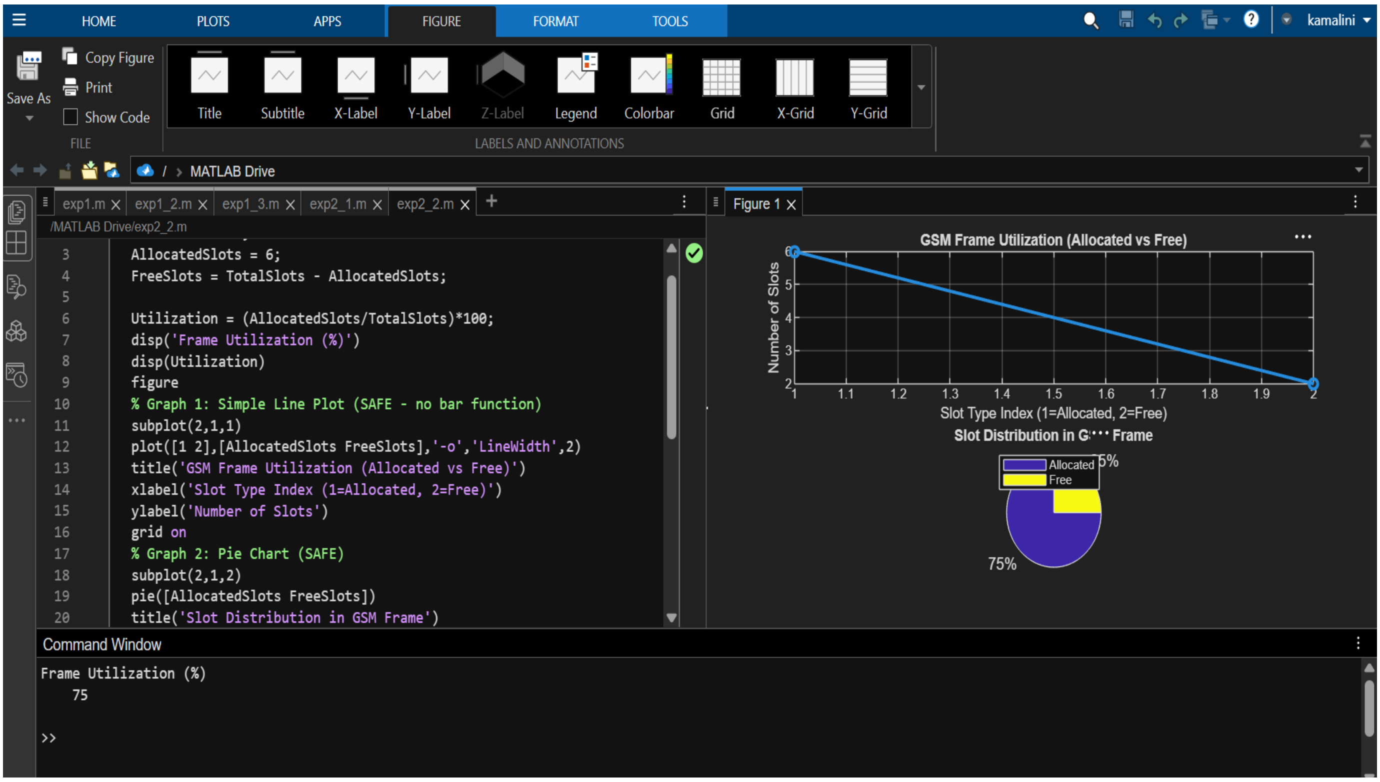
communication.



OBJECTIVE 2 :

To determine the Frame Utilization (%) based on the number of allocated time slots compared to total available slots. This evaluates the efficiency of TDMA resource usage.

```
clc; clear; close all;
TotalSlots = 8;
AllocatedSlots = 6;
FreeSlots = TotalSlots - AllocatedSlots;
Utilization = (AllocatedSlots/TotalSlots)*100;
disp('Frame Utilization (%)')
disp(Utilization)
figure
% Graph 1: Simple Line Plot (SAFE - no bar function)
subplot(2,1,1)
plot([1 2],[AllocatedSlots FreeSlots],'-o','LineWidth',2)
title('GSM Frame Utilization (Allocated vs Free)')
xlabel('Slot Type Index (1=Allocated, 2=Free)')
grid on
ylabel('Number of Slots')
% Graph 2: Pie Chart (SAFE)
subplot(2,1,2)
pie([AllocatedSlots FreeSlots])
title('Slot Distribution in GSM Frame')
legend('Allocated','Free')
```



OBJECTIVE 3 :

To compute the Number of Allocated Time Slots for multiple users in the GSM system.

This ensures proper scheduling and interference-free communication.

CODE:

clc; clear; close all;

Users = 1:8; % Total users

TotalSlots = 5; % Available slots

Alloc = zeros(1,8); % Initialize allocation

Alloc(1:TotalSlots) = 1; % Assign slots to first users

Blocked = 1 - Alloc; % Remaining users blocked

disp('User Wise Slot Allocation')

disp(table(Users',Alloc','VariableNames',{'User','Status(1=Allocated,0=Blocked)'})

) figure

% Graph 1: Allocation per user

subplot(2,1,1)

stem(Users,Alloc,'filled','LineWidth',2)

title('TDMA Slot Allocation per User')

xlabel('User Index')

```
ylabel('Slot Status (1=Allocated, 0=Blocked)')

ylim([-0.2 1.2])

grid on

% Graph 2: System view (safe alternative to bar)

subplot(2,1,2)

plot(Users,Alloc,'-o','LineWidth',2)

hold on

plot(Users,Blocked,'-s','LineWidth',2)

title('TDMA Slot Utilization Analysis')

xlabel('User Index')

ylabel('Status')

legend('Allocated','Blocked')

grid on
```